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SPACE



SpaceX worker injury rates at Starbase outpace industry rivals



SpaceX employees are more likely to be injured while working at Starbase than any of its other manufacturing facilities, according to company worker safety records reviewed by TechCrunch.

Starbase is a sprawling launch-and-manufacturing site that recently incorporated as its own Texas city, logged injury rates that were almost 6x higher than the average for comparable space vehicle-manufacturing outfits and nearly 3x higher than aerospace manufacturing as a whole in 2024, according to Occupational Safety and Health Administration (OSHA) data released in May. That outsized injury rate has persisted since 2019, when SpaceX began sharing Starbase injury data with the federal regulator.

IMAGE CREDITS: SPACEX

Starbase is home to SpaceX's most ambitious program: a fully reusable, ultra-heavy-lift rocket called Starship. The company has been moving at a breakneck pace to bring Starship online to launch Starlink internet satellites and other payloads.

Since [Starship's first orbital test](#) in April 2023, SpaceX has attempted eight additional integrated flights. During three of those tests, [the company made history](#) by catching the massive Super Heavy booster with "chopstick" arms attached to the launch tower.

The data suggests that SpaceX's rapid progress comes at a cost. And while injury rates alone don't

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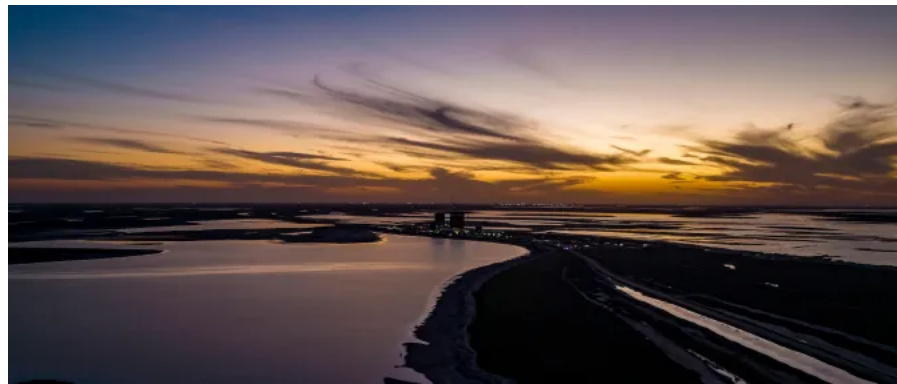
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provide a complete picture of the safety culture at Starbase, they do offer a rare glimpse into the working conditions of the world's leading space company.

Breaking down Starbase numbers



STARBASE CITY, AN UNINCORPORATED TOWN IN TEXAS.

IMAGE CREDITS:SPACEX

OSHA uses a standardized safety metric called Total Recordable Incident Rate (TRIR) to measure a company's safety record and compare it to industry peers, like Blue Origin and United Launch Alliance. The publicly available data has limitations. It doesn't distinguish between minor injuries like stitches versus serious incidents such as amputations.

TechCrunch calculated the TRIR based on that data, which includes the total number of incidents and total number of hours worked by SpaceX employees at each site.

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Starbase, which plays a central role in SpaceX CEO Elon Musk’s mission to make life multi-planetary, is an outlier in the company and across the industry as a whole. Its TRIR topped out at 4.27 injuries per 100 workers in 2024, when it employed an average of 2,690 workers, according to the data submitted to OSHA. Injured Starbase employees were unable to perform their normal job duties for a total of 3,558 restricted-duty days, plus 656 lost-time days where injuries made them unable to work at all.

Starbase is classified by the U.S. government as a space vehicle-manufacturing operation. The injury rate in this sector has fallen dramatically [since 1994](#), dropping from 4.2 injuries per 100 workers to [0.7 injuries per 100 workers in 2023](#), according to historical data from the Bureau of Labor Statistics. (BLS calculates these rates through its annual company surveys, which asks for the same information found in OSHA’s worker injury forms.) But despite major changes in safety processes across the industry, Starbase is closer to the rates of 30 years ago.

The injury rate across all of SpaceX’s manufacturing facilities — which includes an engine development and testing site in McGregor, Texas; a Starlink satellite-manufacturing complex in Bastrop, Texas; the Falcon rocket complex in Hawthorne, California; and another satellite-

manufacturing site in Redmond, Washington — is 2.28.

These other facilities report lower TRIR rates, though most still exceed the industry averages. For instance, 2024 data shows TRIR rates of 2.48 at McGregor, 3.49 at Bastrop, 1.43 at Hawthorne, 2.89 at Redmond. The 2024 TRIR for aerospace manufacturing as a whole is 1.6.

SpaceX also operates several non-manufacturing sites, including barge operations off both coasts; offices in Sunnyvale, California; and launch sites at Cape Canaveral and Vandenberg Space Force Base.

Former OSHA chief of staff Debbie Berkowitz told TechCrunch via email that Starbase’s TRIR “is a red flag that there are serious safety issues that need to be addressed.”

However, there is a debate among safety professionals about whether TRIR is the most reliable metric for assessing and predicting injury rates, particularly serious incidents like fatalities, and especially for small companies. [A recent paper](#) on TRIR questioned its statistical validity and advocated that organizations use alternative measures of safety performance instead.

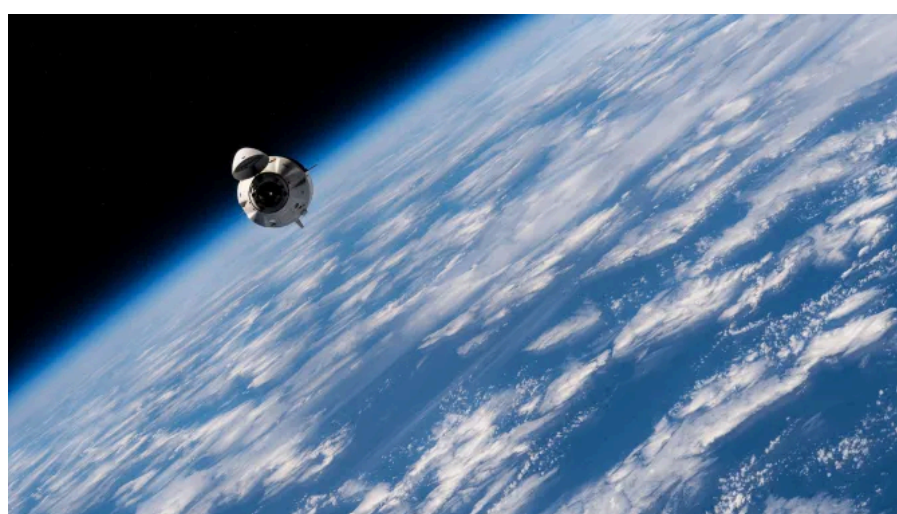
Of the 14 OSHA inspections at SpaceX facilities over the past four years, six involved accidents and injuries at Starbase. That includes a partial finger amputation in 2021 and a crane collapse in June 2025. The latter inspection is still ongoing. Investigations by other news outlets, [including Reuters](#), have uncovered hundreds of previously unreported worker injuries, including crushed limbs and one fatality.

The 2024 injury rate at Starbase marks an improvement to that of the prior year, which topped

out at 5.9 injuries per 100 workers in 2023 and 4.8 injuries in 2022. But it still leads among SpaceX's land-based facilities and is second overall only to its West Coast booster recovery operations, which has a TRIR of 7.6.

OSHA confirmed TechCrunch's calculation of Starbase's TRIR over email, but otherwise did not respond to questions regarding that location's injury rate. SpaceX did not respond to request for comment.

NASA's stake



NASA CREW-2 MISSION IN 2021 RETURNS TO EARTH.

IMAGE CREDITS:SPACEX UNDER A CC BY NC 2.0 LICENSE.

NASA has a major stake in Starship's development. The agency is counting on using the rocket to return humans to the moon before the end of this decade, and it is [paying more than \\$4 billion](#) to SpaceX for two crewed Starship flights to the lunar surface.

Both the contract for the [Starship lander](#) and SpaceX's contract for its Commercial Crew services to the International Space Station contain particular clauses that allow the agency to take action in the case of a major breach of safety, such as a fatality or a "willful" or "repeat" OSHA violation.

While a persistently high TRIR rate can be evidence of a safety problem, it is not an automatic trigger for

action and does not fall [under the definition](#) of a “major breach of safety” in their contracts.

“NASA interacts frequently with its partners, including SpaceX, to ensure safety from a mission assurance perspective, and remains in regular contact with the company during normal contract administration,” a NASA spokesperson told TechCrunch in response to questions about the company’s TRIR. “Safety is paramount to NASA’s mission success. The agency continues to work with all our commercial partners to build and maintain a healthy safety culture.”

Among rocket makers with vehicles in operation, Starbase still leads the pack: at ULA’s manufacturing facility in Decatur, Alabama, the TRIR is 1.12 injuries per 100 workers; at Blue Origin’s rocket park on the coast of Florida, the rate is 1.09.

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Aria Alamalhodaie

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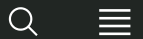
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IMAGE CREDITS:

A crane collapsed at SpaceX's South Texas rocket facility this week, and the company's newly formed city won't say if anyone was hurt.

On June 23, a crane being used to clean up debris from the most recent SpaceX rocket explosion collapsed at the company's launch complex. Footage of the accident was [captured by Lab Padre](#), one of the content creators who film and photograph the site on a regular basis. But it was filmed from far away, making it impossible to tell whether anyone was harmed or in danger.

SpaceX has not publicly acknowledged the collapse and did not respond to TechCrunch's request for comment. That's not surprising; while the company posts details about spaceflight

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mishaps, like when its Starship rockets blow up, it is otherwise tight-lipped.

But SpaceX’s facility is now part of a newly incorporated city called Starbase, Texas, and this accident happened within its borders. And after one of the Starship rockets [blew up on a test stand](#) last week, Starbase officials [made a short post on X about it](#).

There has been no post about the crane collapse, though, and the city has ignored direct appeals for information. TechCrunch has contacted Starbase’s main media email address, its mayor, its two commissioners, its city administrator, and its clerk this week. None have responded to multiple requests for information about the accident.

The collapse is one of the first tests of whether Starbase city has an appetite (and will) for transparency while being run by a SpaceX executive, and acting as a home for the company and more than 200 of its employees. Starbase is also financially beholden to Elon Musk’s spaceflight company: Just this week, the city [finalized](#) a \$1.5 million loan from SpaceX to fund the city’s operations through September 2025.

The idea of making a city called Starbase was first [floated by Musk in 2021](#). But it wasn’t until May of this year that it formally became a city. Its citizens — mostly SpaceX employees — voted overwhelmingly to incorporate. SpaceX’s vice president of “Texas Test and Launch,” Bobby Peden, became mayor. The company’s senior director of environmental, health, and safety, Jordan Buss, became a commissioner. The other commissioner, Jenna Petrzelka, worked at SpaceX for years.

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Those officials have been busy shaping the new city in the last few weeks. The city sent out a [notice](#) in late May to residents in a newly proposed “mixed use district” that they may “lose the right to continue using” their property. The city also [started erecting gates at SpaceX's request](#) that limit access to the city, citing safety concerns. Buss said in a meeting that Starbase would grant access to outsiders “if there’s a need to be in the city” and provide “access codes” to ambulances, firefighters, and law enforcement, according to [Valley Central](#).

It's not clear if those kinds of emergency services were needed Tuesday when the crane collapsed. The Cameron County Sheriff's Department told TechCrunch that it did not receive any calls from Starbase around the time of the accident. Local fire officials from the nearby city of Brownsville and from Cameron County did not respond to requests for comment.

All we know about the collapse — aside from seeing it happen — is that the Occupational Safety and

Health Administration (OSHA) has [opened an investigation](#), according to CNBC.

SpaceX has a troubled [history of worker safety](#). But OSHA investigations take time, and the agency has been diminished by cuts [from Musk's DOGE project](#). Musk's companies also do not have very constructive relationships with OSHA. In 2019, Tesla [refused to let OSHA inspectors enter its factory in Nevada](#), even when those inspectors returned with a sheriff's deputy and a warrant signed by the judge.

Topics: [Elon Musk](#) [Space](#) [SpaceX](#)



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Sean O'Kane is a reporter who has spent a decade covering the rapidly-evolving business and technology o...

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SpaceX launches mammoth Starship rocket and brings it back for the first time

Aria Alamalhodaei — 7:33 AM PDT · June 6, 2024

[SpaceX](#) has shown once again that subjecting rocket technology to the real-world flight environment pays dividends. In its latest launch, the company achieved a key milestone in its Starship flight test campaign: returning the booster and the upper stage back to Earth in controlled ocean splashdowns.

Bringing the two parts of the launch vehicle back — the Super Heavy booster and the upper stage, which is also called Starship — are essential to the company's long-term plans to make Starship the first-ever fully reusable rocket. But to reuse, you have to recover, and SpaceX is proving that it will be able to do just that with Starship.

The ultimate goal is to fly Super Heavy and the Starship upper stage back to Starbase, SpaceX's private Starship launch and development site in southeast Texas, where they'd make vertical landings on solid ground. A controlled ocean splashdown is the first step toward executing this plan. SpaceX was the first company to ever reuse a part of a rocket that's flown to space, but even its workhorse Falcon 9 rocket is only partially reusable (the second stage is expended in orbit).

IMAGE CREDITS: [SPACEX](#)

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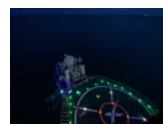
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SpaceX faces two new lawsuits alleging safety-related retaliation

Aria Alamalhodaei
- Jul 30, 2025

Starship lifted off from Starbase at 8:50 a.m. CT, the fourth launch in the rocket test campaign. SpaceX aims to use the rocket to launch heavier versions of its Starlink satellites, deliver astronauts to the moon for NASA, and eventually make life multi-planetary.

Although this is only the fourth time that the mammoth Starship has ever launched to orbit, the test went remarkably smoothly. Only one of the 33 Raptor engines on the Super Heavy booster went out, and the company completed its novel “hot-staging” stage separation technique. (During hot staging, the engines on the upper stage ignite temporarily while it is still attached to the booster, essentially helping to “push” the booster away.) SpaceX also successfully jettisoned the “hot stage ring” that sits in between Starship and Super Heavy for the first time, in order to reduce the weight of the booster as it returns to Earth.

Shortly after launch, the booster successfully splashed down in the Gulf of Mexico. A little over an hour after launch, Starship followed suit, surviving the extreme heat from traveling through the Earth’s atmosphere at hypersonic speeds and splashing down in the Indian Ocean.

Super Heavy has splashed down in the Gulf of Mexico pic.twitter.com/hlY3Gkq57k

— SpaceX (@SpaceX) [June 6, 2024](#)

Of the 18,000 heat shield tiles on the ship, engineers intentionally replaced one with a thinner version and removed two tiles completely in order to “measure how hot things get without tiles in those locations, while also testing some thermal protection options,” the company said. SpaceX CEO [Elon Musk later said on X](#) that the ship successfully splashed down despite having a damaged flap and losing “many tiles.” Musk has said before that the

heat shield system is [“the biggest remaining problem”](#) in Starship development.

Starship has come a long way since [its first orbital test flight in April 2023](#) — that launch saw the mid-air explosion of both parts of the rocket and the dysfunction of many of the rocket engines. Each subsequent test has gone further, and Starship reached orbit for the first time in the third test this past March. During that test, SpaceX also tested capabilities that will be key for delivering payload to space, including opening and closing the payload door.

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